

NEXORA CARE FLOW

Appointment Records – Data Quality & Cleaning Report

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1. Purpose of the Data Cleaning Exercise

As part of the Nexora Care Flow project, the Appointment Records dataset was reviewed and prepared for analysis and future appointment forecasting.

The purpose of the exercise was to ensure that the information being used to support healthcare planning and decision-making was **complete, consistent and reliable**. The review focused on missing information, duplicate appointments, appointment dates and times, and the consistency of appointment statuses.

The dataset contains **129,353 appointment records covering 15 areas of information**, including patient, clinic, provider, appointment, booking and holiday information.

2. Key Data Quality Findings

Missing Holiday Information

The main issue identified was in the **Holiday Name** field. Of the 129,353 appointment records, **128,414 did not contain a holiday name**, while 939 did.

This was investigated against the holiday indicator. The missing information was found to relate to appointments that were not recorded as taking place on a holiday. Rather than removing these appointments, they were labelled "**No Holiday**".

This was an important decision because removing these records would have resulted in the loss of a large proportion of the available appointment information.

Outcome: The holiday information is now represented consistently and all appointment records have been retained.

Duplicate Appointments

The dataset was checked for records that appeared more than once.

Finding: No exact duplicate records were identified.

This means there was no evidence from the duplicate check that appointment volumes were being artificially increased by repeated records.

3. Appointment Date and Time

The appointment date and time information was reviewed because accurate timing is essential for understanding patterns in healthcare demand.

The date and time information was standardised into a consistent format and checked for missing information.

Finding: All 129,353 appointment records contained an appointment date and time.

This provides a reliable basis for examining when demand occurs, including differences between days, weeks, months, seasons and holiday periods.

Why this matters

Reliable appointment timing will allow the project to identify patterns in demand and support the development of a forecasting approach that can help stakeholders anticipate future appointment requirements.

4. Appointment Status Review

Appointment outcomes were reviewed to identify any unusual or inconsistent entries.

Four appointment statuses were identified:

Appointment Status Number of Records

Completed	109,040
No-Show	11,367
Cancelled	6,296
Rescheduled	2,650
Total	129,353

The review found **four clear and consistent appointment outcomes**. No additional or obviously incorrect status categories were identified.

The largest group was **Completed appointments**, while **Rescheduled appointments** represented the smallest group.

Why this matters

Having consistent appointment outcomes allows the project to accurately assess appointment utilisation, cancellations, no-shows and rescheduling patterns. These measures can provide useful information for future service planning.

5. Review of Potential Data Anomalies

The data was also reviewed for potentially unusual records, including inappropriate booking lead times and other values that could affect the reliability of the analysis.

Records with potentially unusual values were investigated rather than automatically deleted. This approach helps preserve the original information while ensuring that questionable records can be reviewed before being used for decision-making.

Clinic appointment volumes were also considered as part of the data-quality review. Unusually low activity should not automatically be interpreted as a clinic closure, as there may be other reasons for a low number of appointments. Further confirmation from operational records would therefore be appropriate before making such a classification.

6. Overall Outcome

The cleaning exercise has produced a more consistent and reliable appointment dataset for the next stage of the Nexora Care Flow project.

Summary of outcomes:

Area Reviewed	Outcome
Appointment records	129,353 records reviewed
Missing holiday information	Appropriately labelled " No Holiday "
Duplicate records	None identified
Appointment date/time	Standardised and complete
Appointment statuses	Four consistent categories identified
Potential anomalies	Reviewed and retained for appropriate validation
Cleaned dataset	Prepared for further analysis

7. Stakeholder Implications

The cleaning exercise provides greater confidence that the appointment data can be used for meaningful analysis.

In particular, the work ensures that:

- Valid appointment records have not been unnecessarily removed.
- Holiday information is represented consistently.
- Appointment dates and times can be reliably analysed.
- Appointment outcomes can be compared consistently.
- Duplicate records are not contributing to appointment volumes.
- Potential data-quality issues can be identified before they influence forecasting results.

The cleaned dataset therefore provides a stronger foundation for the next stage of **Nexora Care Flow**, where appointment patterns and demand will be analysed to support better planning and forecasting.

8. Conclusion

The Appointment Records dataset has been successfully reviewed and prepared for the next stage of the Nexora Care Flow project. The review identified and addressed the principal data-quality issue relating to holiday information, confirmed that no exact duplicate records were present, standardised appointment date and time information, and confirmed four consistent appointment outcomes.

The resulting dataset is now suitable for exploratory analysis and the development of appointment-demand forecasting, with the aim of providing stakeholders with clearer insight into future healthcare appointment requirements.